

monotropism-comorbidity-landscape

Monotropism and Its Web of Comorbidity: Direct and Indirect Links to Behavioural, Psychiatric, and Neurodevelopmental Conditions

A cited research brief.

Executive summary

Monotropism (Murray, Lawson & Lesser, 2005) is a theory of autistic attention that frames autism as a difference in *resource allocation*: a monotropic mind concentrates processing resources into a small number of highly-aroused "attention tunnels," leaving fewer resources for peripheral channels. A 2023 self-report measure — the **Monotropism Questionnaire (MQ)** — provides the first empirical validation, showing autistic and ADHD people score significantly higher than non-autistic/non-ADHD controls [1][2][3]. A closely related peer-reviewed model, Goldknopf's **atypical resource allocation** account, argues the same narrowed-intense attention can tie together autism's sensory, social, executive, and motor features — while cautioning it is "most likely a factor in autistic people with strong sensory symptoms," not the sole cause [4].

Because monotropism is a theory of *how resources are spent*, it plausibly ripples into many domains. The honest picture, however, splits into three tiers:

- **Tier 1 — Direct, empirically supported links.** The autism–ADHD overlap ("AuDHD") is the strongest: the MQ shows monotropic attention is elevated in *both* conditions, reframing the "monotropic split" (competing or unstable tunnels) as a real, measurable

attentional phenotype [2][3][5]. Mechanism-level overlaps — atypical **interoception** and **executive dysfunction** — are well evidenced and help explain downstream problems with eating, clutter, and avoidance [6][7][8].

- **Tier 2 — Indirect, shared-mechanism comorbidity.** Autism overlaps markedly with **anorexia/restrictive eating** and **OCD**. In each case the comorbidity is real and replicated, but the *best-evidenced* bridge is a shared mechanism (interoception; ego-syntonic vs ego-dystonic processing) rather than monotropism specifically [6][9][10].
- **Tier 3 — Plausible extrapolation or contested constructs.** Some links in the popular framing — "doom piles" as a monotropic object-permanence failure, agoraphobia as a primary monotropic outcome, **PDA** as a distinct profile — rest on weaker or contested evidence and should be treated as hypotheses, not established facts [11][12][13].

Throughout, a recurring theme: many behaviours that look like separate "disorders" are better read as **the cognitive cost of a narrow attention system colliding with a polytropic world** — but only some of those readings have been directly tested.

1. Foundations: what monotropism is, and how solid it is

Monotropism proposes that minds differ in how they distribute attentional resources. A **polytropic** mind keeps many interest channels active at once (broad, shallow, easy switching); a **monotrophic** mind funnels resources into a few intensely-aroused "attention tunnels" (narrow, deep, hard to switch) [1][3]. The originators — Dinah Murray, Wenn Lawson and Mike Lesser — first presented the idea at the Durham autism conferences from 1992 and formalised it in 2005, arguing it explains autistic traits (special interests, need for sameness, distress at transitions, detail-focus) better than deficit-only models [1][14].

Its scientific status has shifted in the last few years:

- It is **empirically measurable**. The MQ (Garau et al., 2023) is a 47-item self-report instrument tested on 1,110 participants (756 autistic, 354 non-autistic). Autistic and ADHD participants scored significantly higher than controls [2][3].

- It has a **peer-reviewed mechanistic cousin**. Goldknopf's "atypical resource allocation" model (*Frontiers in Integrative Neuroscience*, 2013) operationalises the intuition: in autism, attentional resources are **narrowed** (to fewer/smaller areas, modalities, and processing stages, sometimes more intensely) and/or **reduced** in overall capacity — a framework that explicitly ties together language, social cognition, sensory/attentional atypicality, and executive/motor differences [4].
- It sits **alongside, not instead of**, other cognitive theories (Theory of Mind; weak central coherence / detail-focused style (Happé & Frith, 2006); executive dysfunction) [4][15].
- It is still a **model, not a biomarker**. Goldknopf is explicit that it is one factor among several, "most likely" salient in autistic people with strong sensory symptoms [4]. Murray and colleagues similarly frame monotropism as a unifying *lens*, not a complete aetiology [1][14].

Implication for the rest of this brief: monotropism is a credible, increasingly-validated organising idea, but its *explanatory reach* into each comorbidity must be argued case by case — not assumed.

2. Tier 1 — Direct links: the neurodevelopmental cluster

2.1 AuDHD and the "monotropic split" (*strongest direct link*)

Autism and ADHD co-occur far more often than chance would predict: ADHD is present in roughly **30–80%** of autistic people, and autism in roughly **20–50%** of those with ADHD [5]; a meta-analysis puts ASD-symptom rates in ADHD samples around **21%** [16]; one large Medicaid study found **27%** of autistic adults without intellectual disability had a co-occurring ADHD diagnosis — about 10× the general rate [17]. The overlap is substantially genetic (~50–72%) [18].

Monotropism helps resolve an old puzzle: if autism is "hyper-focus" and ADHD is "inattention," why do they co-occur? The MQ answers this empirically — **ADHD people, especially AuDHD people, are also highly monotropic** [2][3]. Hyperfocus in ADHD (Grotewiel et al., 2022) is conceptually near-identical to the monotropic tunnel [19]. The "monotropic split" framing therefore re-casts AuDHD not as opposites but as **an unstable system of competing or rapidly-shifting attention**

tunnels: the brain demands deep monotropic focus but cannot stabilise the tunnel, producing the familiar AuDHD oscillation between intense engagement, rapid burnout, and chronic under-stimulation [19][20].

Caveat: the "multiple competing tunnels" mechanism is *plausible and consistent with the MQ data* but not yet directly tested as a dynamic; it should be read as a strong hypothesis.

2.2 Autistic burnout and tunnel disruption

Autistic burnout — chronic exhaustion, loss of skills, and reduced tolerance — is increasingly linked to the cumulative cost of constant tunnel-switching, masking, and demands exceeding capacity, with insufficient recovery/flow time [20][21]. This is an area where monotropism is heuristically powerful (every forced transition is a tunnel rupture), though the burnout literature itself is still maturing and the monotropism→burnout pathway is largely clinical/first-person rather than experimentally established [20].

2.3 Demand avoidance and PDA (*contested construct*)

A demand — "get ready," "do this form" — can be experienced as a violent intrusion into a flow state. This is the intuitive monotropic account of **demand avoidance**. The more specific construct **Pathological Demand Avoidance (PDA)**, coined by Elizabeth Newson in the 1980s and now often reframed as "Pervasive Drive for Autonomy," describes an anxiety-driven need for control and avoidance of ordinary demands [11][22].

Two things must be separated:

- **Demand-avoidant behaviour is real and common** in autism, and the monotropic "tunnel disruption" account is a reasonable *mechanistic story* for the panic and resistance it produces.
- **PDA as a *distinct profile or subtype* is not established.** The UK's National Autistic Society states "no research has found strong evidence for [PDA] as a distinct profile" [11]; Newson's foundational study has been criticised for **circular reasoning and over-interpretation** (groups defined by the researchers were then "studied") [12][13]; and reviews note persistent problems with definition, construct validity, and measurement [23]. A 2023 review is

pointedly titled "*Pathological Demand Avoidance: Current State of Research and Critical Discussion*" [13].

So the seed text's framing is half-right: the *experience* (a demand shatters the tunnel) is well-described by monotropism; the *diagnostic entity* (PDA) is contested and should not be presented as settled.

3. Tier 2 — Indirect, shared-mechanism comorbidity

3.1 Anorexia and restrictive eating

The autism–anorexia overlap is robust and bidirectional: roughly **20–30%** of eating-disorder patients show clinically significant autistic traits, with the highest rates in anorexia nervosa and ARFID [6][24]; conversely, autistic adults have elevated eating-disorder rates. Critically, autistic traits are observed **before** ED onset, so the overlap is not merely a starvation effect [6].

The best-evidenced *mechanistic* bridge is **interoception** — the perception of internal bodily signals (heartbeat, satiety). Atypical interoception is implicated in the aetiology of *both* autism and eating disorders and "may, at least in part, be responsible" for their comorbidity [6]. The same interoceptive atypicality underlies **alexithymia** (difficulty identifying/verbalising emotions), present in ~50% of autistic people and itself tied to eating pathology [7][8].

Where monotropism fits (and doesn't): it is reasonable to say that a monotropic person deep in a tunnel may *miss* hunger cues — but the seed text's specific claim that "monotropism reduces interoception" overstates the causal chain. The evidenced relationship is **autism** → **atypical interoception/alexithymia** → **eating vulnerability**; monotropism is a parallel feature of autism, not the demonstrated *cause* of the interoceptive difference [6][7]. Food/calorie-tracking can also become a tunnel in its own right, but this is mechanism-plausible rather than tested.

3.2 OCD vs hyper-focus: a genuine distinction

Autism and OCD co-occur substantially: ~**17%** of autistic people meet OCD criteria, versus ~1–2% in the general population [9]. Despite surface similarity (repetitive behaviour, rigidity), the **affective driver differs**

and this is well-established clinically:

- **Autistic** restricted/repetitive interests and routines are typically **ego-syntonic** — enjoyable, self-soothing, interest-driven, often regulating.
- **OCD** compulsions are **ego-dystonic** — unwanted, anxiety-driven, performed to neutralise feared outcomes [9][10].

The seed text's distinction is therefore correct and evidence-backed, and is confirmed directly by autistic participants themselves: in a qualitative study they reported that repetitive behaviours are "part of who they are," whereas OCD symptoms "conflicted with how they view themselves" [10]. The same study aptly frames the co-occurrence as "*Autism is the arena and OCD is the lion*" — OCD symptoms can "hijack" autistic special interests, inflating their intensity and distress [10]. Monotropism adds context (a tunnel can be *captured* by OCD content) but the differentiation rests on the ego-syntonic/dystonic axis, not on monotropism per se.

3.3 Anxiety and affective disorders (the substrate under "agoraphobia")

Anxiety is among the most common autism comorbidities (one secondary review cites ≥ 1 psychiatric comorbidity in ~86% of autistic people) [25]. Two mechanism points matter for the behavioural cluster below: (i) **sensory over-responsivity** and anxiety are linked, and through *context conditioning* produce **behavioural avoidance of generalised locations** (restaurants, shops, parties) [26]; (ii) the frequent co-occurrence of alexithymia/interoceptive difficulty means emotional states are often poorly labelled, fuelling diffuse anxiety [7].

4. Tier 3 — Behavioural/functional cluster: clutter, hoarding, avoidance, social withdrawal

4.1 Clutter, "doom piles," and executive dysfunction

The *phenomenon* is well-supported: autistic and ADHD people show high rates of disorganisation and clutter, and the best-evidenced route is **executive dysfunction** — deficits in categorisation, decision-making, planning, and working memory [8][27]. Hoarding symptoms are elevated

in both ADHD (with a "special link with inattention" [8a]) and autism [8]; in autism specifically, qualitative work shows difficulty discarding is tied to **interest attachments and meaning**, not merely object value [28].

Caveat on the seed text: the specific claim that clutter persists because of an autistic "**object-permanence deficit**" ("out of sight, out of mind") is **not supported by peer-reviewed evidence** I could locate. Object permanence is typically *intact or superior* in autism; the "out of sight, out of mind" experience is better read as a **working-memory/attention-allocation** effect (a tunneled mind does not re-orient to stored items) than a perceptual-object-permanence failure. The evidenced mechanism for clutter remains **executive dysfunction plus interest-attachment**, which overlaps the monotropic/AuDHD population but is not uniquely "monotropic" [4][8][28].

4.2 Hoarding-spectrum behaviour

Hoarding disorder affects ~1.5–6% of the general population but is elevated in autism [29]. In autistic adults, hoarding-like difficulty discarding is frequently reported alongside **cognitive (executive) difficulties and strong interest/meaning attachments** — items are nodes in a web of ideas and future utility, so discarding feels lossy [28]. This is consistent with a monotropic *style* (deep processing of objects) but, again, the evidenced driver is EF/attachment, shared across neurodevelopmental conditions [8].

4.3 Agoraphobia and "safe-space" bounding (*needs correcting*)

The seed text frames agoraphobic home-confinement as a near-direct monotropic outcome. The data are more nuanced:

- Agoraphobia is elevated in autism — ~**23–25%** of autistic adults (without intellectual disability) vs ~1.3% in the general population — **but it is less common than social phobia or GAD**, and the effect concentrates in those without ID [25].
- The better-evidenced driver is **sensory over-responsivity + social anxiety**, with context-conditioned avoidance of overwhelming locations — i.e., the "safe-space" behaviour is real but is **secondary to sensory/social anxiety**, not a primary monotropic phenomenon [25] [26].

So: monotropism helps explain *why* the outside world feels disruptive (unpredictable, polytropic, tunnel-shattering), but classifying the resulting confinement as "agoraphobia" specific to monotropism overstates the link. It is a sensory-anxiety-driven avoidance that *overlaps* the monotropic experience.

4.4 Social withdrawal, info-dumping, and the Double Empathy Problem

Social interaction is genuinely polytropic — it demands simultaneous tracking of faces, tone, hierarchy, timing, and unwritten rules — so it is disproportionately costly for a monotropic system, a point consistent with Goldknopf's resource model [4]. The seed text's intuition is sound.

But the sharper, *empirically supported* frame for the resulting communication breakdown is the **Double Empathy Problem** (Milton, 2012; 2017): the difficulty is *bidirectional* and relational — a "disjuncture in reciprocity between two differently disposed social actors" — not a one-way empathy deficit in the autistic person [30]. Empirical work now shows non-autistic observers both *detect and demonstrate* the double empathy problem in mixed interactions [31]. "Info-dumping" fits naturally: it is depth-seeking communication from a monotropic system, which fails when the partner cannot match the depth — a mismatch of processing styles, not a lack of empathy [30][31].

Gap to flag: the DEP literature still largely excludes autistic children, nonspeaking people, and people with intellectual disability, so generalisation is limited [32].

5. The boundary: where the monotropic account reaches too far

Claim in the seed/lay framing	Evidence status	Better framing
Monotropism <i>causes</i> autism's features	Partly supported (MQ; Goldknopf)	One validated organising factor among several; strongest in sensory-present autism [1][4]

Claim in the seed/lay framing	Evidence status	Better framing
AuDHD = "monotropic split" (competing tunnels)	Strong + directly measurable (MQ)	A real attentional phenotype; dynamic mechanism is hypothesis-grade [2][3]
PDA is a distinct autism profile	Contested	Demand avoidance is real; PDA-as-entity lacks construct validity [11][12][13]
Monotropism <i>reduces interoception</i> → anorexia	Indirect	Bridge is autism→interoception/alexithymia→ED vulnerability; monotropism is parallel, not the shown cause [6][7]
Clutter from an "object-permanence deficit"	Unsupported	Executive dysfunction + interest-attachment; no perceptual object-permanence deficit evidenced [4][8][28]
Agoraphobia as a primary monotropic outcome	Overstated	Secondary to sensory over-responsivity + social anxiety; less common than GAD/social phobia [25][26]
OCD vs monotropic "obsessions" differ by affect	Well-supported	Ego-dystonic (OCD) vs ego-syntonic (autistic interest); OCD can "hijack" tunnels [9][10]
Double Empathy Problem explains social breakdown	Supported (but gaps)	Bidirectional mismatch; samples skew to speaking adults without ID [30][31][32]

6. Open questions

- 1. Dynamic testing of the tunnel.** The MQ is a *trait* measure. Is the "monotropic split" (competing/unstable tunnels in AuDHD) observable in real-time attention/EEG? Largely untested.
- 2. Causal arrow to interoception.** Does a monotropic resource distribution *produce* interoceptive atypicality, or do they share a common upstream cause? Current evidence can't separate these [6][7].

3. **PDA resolution.** Will demand-avoidance research converge on a validated, measurable construct, or remain a contested lay/clinical category? [11][13]
 4. **Clutter mechanism.** Is "out of sight, out of mind" a working-memory/allocation effect (plausible) or something else? No dedicated study found.
 5. **DEP generalisability.** How does the double empathy problem operate for nonspeaking autistic people, children, and people with co-occurring intellectual disability? [32]
 6. **Burnout.** What is the specific, testable contribution of monotropic tunnel-disruption to autistic burnout versus masking, trauma, and chronic demand? [20]
-

Sources

- [1] Murray, D., Lesser, M., & Lawson, W. (2005). Attention, monotropism and the diagnostic criteria for autism. *Autism*.
<https://journals.sagepub.com/doi/abs/10.1177/1362361305051398> [2] Garau, V., Woods, R., Chown, N., Hallett, S., Murray, F., Wood, R., Murray, A., & Fletcher-Watson, S. (2023). The Monotropism Questionnaire (MQ). Open Science Framework (preprint). Commentary/summary:
<https://ndconnection.co.uk/blog/monotropism-and-the-monotropism-questionnaire> ; mirror: <https://stimpunks.org/monotropism-questionnaire> [3] Autistic Realms / Neurodiverse Connection (Edgar) — summary of MQ findings (47 items; N=1,110; autistic + ADHD score significantly higher): <https://autisticrealms.com/monotropism-vs-polytropism-adhd-audhd-autistic-attention> [4] Goldknopf, E. J. (2013). Atypical resource allocation may contribute to many aspects of autism. *Frontiers in Integrative Neuroscience*. PMC3872719.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3872719> [5] Psych Scene Hub — ADHD and autism comorbidity review (ADHD in ~30–80% of ASD; ASD in ~20–50% of ADHD; cites Lau-Zhu 2019; Hollingdale 2019):
<https://psychscenehub.com/psychinsights/adhd-and-autism-comorbidity-a-comprehensive-review> [6] Adams, K. L., Murphy, J., Catmur, C., & Bird, G. (2022). The role of interoception in the overlap between eating disorders and autism: methodological considerations. *European Eating Disorders Review*. PMC9543236.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9543236> [7] Klein, M., Witthöft, M., & Jungmann, S. M. (2025). Interoception in individuals with autism spectrum disorder: a systematic literature review and meta-analysis. *Frontiers in Psychiatry* (10.3389/fpsyt.2025.1573263) — ~50% of autistic people have alexithymia, mediated by interoceptive deficits. <https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2025.1573263/full> [8] Morein-Zamir, S., Kasese, M., Chamberlain, S. R., & Trachtenberg, M. (2020). Elevated levels of hoarding in ADHD: a special link with inattention. *Journal of Psychiatric Research*. PMC7612156. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7612156> — [8a] the "special link with inattention" is the paper's own subtitle. See also Drake Institute on ADHD/executive dysfunction: <https://drakeinstitute.com/articles/adhd/adhd-and-executive-dysfunction-connection> [9] Animosanopsychiatry — systematic-review summary: ~17% of autistic people meet OCD criteria vs ~1–2% general; ego-syntonic vs ego-dystonic distinction: <https://animosanopsychiatry.com/blog/ocd-vs-autism-similarities-differences-and-overlaps> [10] Long, H., Cooper, K., & Russell, A. (2024). "Autism is the arena and OCD is the lion": autistic adults' experiences of co-occurring OCD and restricted/repetitive behaviours (qualitative). PMC11497754. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11497754> [11] National Autistic Society — Demand avoidance ("no research has found strong evidence for [PDA] as a distinct profile"): <https://www.autism.org.uk/advice-and-guidance/behaviour/demand-avoidance> [12] Frontiers in Education (2024). Methods of studying PDA — critique of Newson's circular reasoning (Green et al. 2018; O'Nions & Eaton 2020): <https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2024.1230011/full> [13] Kildahl, A., et al. (2023). Pathological Demand Avoidance: Current State of Research and Critical Discussion. PubMed 36892327. <https://pubmed.ncbi.nlm.nih.gov/36892327> [14] Murray, F. (Lawson) (2018). Me and Monotropism: a unified theory of autism. *The Psychologist* (BPS). <https://www.bps.org.uk/psychologist/me-and-monotropism-unified-theory-autism> [15] Happé, F., & Frith, U. (2006). The weak coherence account: detail-focused cognitive style in ASD. *J Autism Dev Disord* 36:5–25 (cited via Frontiers 2024 comparative study): <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2024.1348074/full> [16] Hollingdale, J., et al. (2019). Meta-analytic review of ASD symptoms in ADHD (cited via Psych Scene Hub and Autistica):

<https://www.autistica.org.uk/what-is-autism/adhd-and-autism> [17] Children's Hospital of Philadelphia — 27% of autistic adults without ID had co-occurring ADHD (~10× general): <https://www.chop.edu/news/rates-adhd-remain-high-adulthood-among-patients-autism> [18] Behavioral Innovations — genetic overlap (~50–72%) summary: <https://behavioral-innovations.com/blog/the-sudden-rise-of-audhd> [19] Autistic Scholar — "Revisiting monotropism" (AuDHD as competing tunnels; references Grotewiel et al. 2022 on ADHD hyperfocus): <https://www.autisticscholar.com/monotropism> [20] Neurodiverse Connection (Edgar) — Monotropism, young people and autistic burnout: <https://ndconnection.co.uk/blog/monotropism-young-people> ; Autistic Realms — Reclaiming rest: <https://autisticrealms.com/reclaiming-rest-autistic-burnout-monotropism-and-resistance> [21] National Autistic Society — What is monotropism? (flow states; references Heasman et al. 2024 "autistic flow theory"): <https://www.autism.org.uk/learn/knowledge-hub/professional-practice/what-is-monotropism> [22] PDA North America — What is PDA? (Newson, 1980s; "Pervasive Drive for Autonomy" reframing): <https://pdanorthamerica.org/what-is-pda> [23] O'Nions, V., et al. (2016). Identifying features of PDA using the DISCO. PMC4820467. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4820467> [24] Nickel, K., Maier, S., Endres, D., & Joos, A. (2019). Systematic Review: Overlap Between Eating, Autism Spectrum, and Attention-Deficit/Hyperactivity Disorder. *Frontiers in Psychiatry*. PMC6796791 (also cites Westwood, Mandy & Tchanturia 2017; Huke et al. 2013). <https://pmc.ncbi.nlm.nih.gov/articles/PMC6796791> [25] Discussing Psychology (Loker) — agoraphobia in autism ~23–25% (without ID) vs ~1.3% general; less common than social phobia/GAD (secondary source, drawing on Helles et al. 2020 and Lugnegård et al. 2012): <https://www.discussingpsychology.com/agoraphobia-and-autism-are-they-intrinsically-linked> [26] Green, S. A., & Ben-Sasson, A. (2010). Anxiety disorders and sensory over-responsivity in children with ASD: is there a causal relationship? *Journal of Autism and Developmental Disorders* — context conditioning and avoidance. PMC2980623. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2980623> [27] Embrace Autism — Autism and hoarding (executive function, decision-making): <https://embrace-autism.com/autism-and-hoarding> [28] Goldfarb, Y., Zafrani, O., Hedley, D., & Yaari, M. (2021). Autistic adults' subjective

experiences of hoarding and self-injurious behaviors (qualitative). *Autism*. PMC8264636. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8264636>

[29] Abouzed, M., Gabr, A., Elag, K. A., Soliman, M., Elsaadouni, N., Elzahab, N. A., & Barakat, M. (2024). The prevalence, correlates, and clinical implications of hoarding behaviors in high-functioning autism. *Scientific Reports* (s41598-024-75371-8). <https://www.nature.com/articles/s41598-024-75371-8>

[30] Milton, D. (2012). On the ontological status of autism: the 'double empathy problem'. *Disability & Society* 27(6):883–887; Milton (2017) extension. Summary: <https://www.autism.org.uk/learn/knowledge-hub/professional-practice/double-empathy>

[31] Jones, D. R., Botha, M., Ackerman, R. A., & King, K. (2024). Non-autistic observers both detect and demonstrate the double empathy problem when evaluating interactions between autistic and non-autistic adults. *Autism*. PMC11308351. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11308351>

[32] Wikipedia — Double empathy problem (acknowledged research gaps: children, nonspeaking, intellectual disability): https://en.wikipedia.org/wiki/Double_empathy_problem

Note on evidence tiers: Tier 1 = direct, replicated measurement (MQ; AuDHD rates); Tier 2 = replicated comorbidity with an evidenced shared mechanism; Tier 3 = plausible extrapolation, single-source, or contested construct. See §5.